keeping hot air oven at 60° C. Fig.1d and e show homogeneous dispersion and attachment of CNTs to UHMWPE powder.

Compositions with low reinforcement content (0.05 and 0.1 wt. %) were chosen for this study to avoid agglomeration. Following nomenclatures (given in parentheses) would be used for these different compositions throughout the manuscript: pure UHMWPE (PE), UHMWPE-0.05 wt. % of HARC (PE-0.05HARC) and LARC (PE-0.05 LARC), UHMWPE-0.1 wt. % of HARC (PE-0.1HARC) and LARC (PE-0.1LARC). Disk shaped pellets of 20 mm diameter and 5 mm thickness were fabricated by compression molding of the composite powder at a pressure of 300 MPa at room temperature, followed by curing at 160° C for 60 minutes in hot air oven.

## 2.2. Characterization

The literatures highlighted that the crystallinity and degradation behavior relatively influences the thermal and mechanical property of polymer based composites. Hence differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA) was performed using SEIKO 6300 EXTAR TG/DSC instrument. About 10 mg of the test samples was sealed in aluminum pans, followed by heating from 35 to 500° C at a heating rate of 5° C/min under nitrogen atmosphere. Degree of crystallinity was calculated using the following equation [38]:

$$X_C = \frac{\Delta H_C}{\Delta H_{100}} \times 100\% \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(1)}$$

where, $X_C$ is degree of crystallization, $\Delta H_c$ is the heat energy per unit mass of the composite. Enthalpy of fusion, $\Delta H_{100}$, is found to be 289 J/g for a 100 % crystalline UHMWPE [39].

In order to understand the effect of CNTs distribution in the composite matrix and within the wear tracks, field emission scanning electron microscope (FE-SEM, Carl Zeiss Ultra Pluss, Germany) was used.

## 2.3 Mechanical and Tribological Characterization

Nanoindentation test was conducted to measure the mechanical properties of the composite using Hysitron TI950 Triboindenter (Hysitron Inc, USA) equipped with three-sided Berkovich diamond indenter with a tip radius of 100 nm. Indentations were made in the depth controlled mode to achieve same depth of penetration in all the indents. The maximum penetration depth of 150 nm was achieved in10s followed by holding for 2s after attaining the peak displacement. A minimum of 25 indents were taken on each pellet to evaluate the hardness (H) and reduced elastic modulus ($E_r$). $E_r$ is calculated from the slope of the initial portion of unloading curve of load versus displacement plots using Oliver and Pharr method [40].

Ball-on-disc tribometer (TR-201E-M2, DUCOM, Bangalore, India) test was performed to evaluate the friction and wear behaviour of the UHMWPE-CNT composite. One of the intended applications of this UHMWPE based material system is in acetabular cup lining of hip implant. This lining goes through several rubbing action with the femoral head, which can be properly addressed through the macro scale wear studies with the rounded (spherical) surface. That is why ball-on-disc tests were chosen for evaluating tribological behavior of the composite. Commercially available EN-31 forged steel ball (hardness - 63 HRC) with 6 mm diameter was used as counter body. Normal load and speed was used for the tests were 5 N and 200 rpm, respectively. The track diameter was approximately 6 mm and test duration was 2 hour. At least three tests were performed in each composition. Contact profilometer (SJ 400, Mitutoyo, Japan) was used to determine the depth and width of the wear track. At least ten profiles across the track were recorded in each of the three tracks for all the compositions to measure the average width and depth of the wear track. Wear volume (V) of the track was calculated using the following formula.

$$V = 2\pi \cdot r \times A \quad \text{.............................} \ (2)$$

where r is the radius of the wear track; A is the cross sectional area of the wear track obtained by integrating the groove in the profile of the wear track. The variation in wear volume, thus measured, is reported as error bars in the corresponding calculations of wear rate.

Electronic sensor was attached to tribometer to record the continuous frictional force. The co-efficient of friction with respect to sliding distance is obtained from there. The specific wear rate was calculated from the following relationship;

$$W = \frac{V}{P \times S} \quad \text{..............................} \ (3)$$

where W is the specific wear rate; V is the total wear volume; P is the normal load and S sliding distance travelled in 2 hours. The worn surfaces were thoroughly studied through FE-SEM to identify the mechanisms dominating their tribological behavior.

## 3. RESULTS AND DISCUSSION

### 3.1 Microstructural Characterization of the Composite

Mixing of initial powders using probe sonication (power-500 Watts and 20 kHz frequency) is found to be very effective in distributing the CNTs very uniformly with PE powders, irrespective of their significant difference in sizes. Fig 1d and e show CNTs getting homogeneously distributed and attached to the polymer powder surface as a result of probe sonication. This is very important for effective use of the total reinforcement volume and uniform mechanical/tribological behavior of the composite. The curing temperature of the polymer and composites was determined through differential scanning calorimetry (DSC) of the as-received PE powder (Fig.2). The melting temperature of PE is nearly 123° C, but it is very viscous at this point. Whereas, proper curing and wrapping of the reinforcement phase by polymer matrix needs more flow and lesser viscosity. Thus a higher temperature of 160°C was used for curing. This was optimized by trial and error at a range of temperatures and by observing the integrity of cured structure through microscopic investigation at the cross section of the PE pellets. Those results are not included in this study, as they are not found very significant to the objective of this study other than just optimizing the curing temperature.

Fig. 3a and b show the fracture surface of the HARC and LARC reinforced composite with 0.1 wt. % CNT content. CNTs are nicely embedded with polymer matrix in both the composites. Both types of CNTs are found to be deeply rooted in the matrix with protruded ends revealed at fracture surface, which indicates strong interfacial bonding with the polymer matrix resulting in effective load transfer. However, the protruded ends are found to be longer in case of HARCs than in LARCs. The reason could be the lower aspect ratio of the latter, and/or uprooting of the same at higher load, as compared to the former.

### 3.2 Thermal Analysis of the Composites

Fig.4 shows the DSC results of the composite. The peak in the DSC curve corresponds to the melting temperature and area under the peak is equal to the enthalpy of melting. Researchers [41-43] have reported that CNTs can act as nucleating agent for the crystallization of polymer. Addition of a mere 1 wt. % MWCNTs also enhance the rate of crystallization up to a significant distance from their vicinity. As a result, crystallinity of polymer increases without a change in melting temperature of the composite. On the contrary, reduction in crystallinity of polyethylene matrix from 58 to 48 % for 5 wt. % CNT [3] and 32.5 to 27.6 % for 10 wt. % of CNT [44] is also reported by other researchers. However, no study has reported yet the effect of aspect ratio of CNT, if any, on the crystallinity of polymer matrix composite. To understand this, DSC tests were conducted and the results were analyzed. The enthalpy of melting for pure PE, PE-0.05LARC and PE-0.1LARC is 88.1 J/g, 84.5 J/g and 70.3 J/g, respectively (Fig.4). The percentage of crystallinity was calculated by dividing the enthalpy of melting of the prepared composite with the enthalpy of melting for 100 % crystalline PE [38], following equation 1. Crystallinity recorded is 30 %, 29 % and 25 %, respectively, for PE, PE-0.05LARC and PE-0.1LARC. Addition of lower aspect ratio CNTs reduces the crystallinity of the composite to some extent, which can be due to two reasons. Firstly, LARCs are dispersed uniformly in PE matrix as stiff fibers (Fig.1e), which might form tube to tube connection in some cases. This network enhances the thermal conductivity of the composite owing to the excellent thermal conductivity of 3,000 W/m·K [45] for CNTs. As a result a net increase in the cooling rate of the composite during solidification is attained, leaving less time for arrangement of the polymer chains to form crystallites. This type of behavior is also observed by other researchers [3, 46]. Secondly, LARCs have higher diameter as compared to polymer chains and hence they act as obstacles and restrict the movement of chains during solidification. This may be other reason for reduction in the crystallinity of the LARC reinforced composites. Agglomeration of nanotubes could have been another reason for decrease in crystallinity. However, significant agglomeration of nanotubes was not observed in the fracture surface of the composites (Fig. 3) in the present study.

PE-0.05HARCs show almost similar crystallinity of 29.76% as PE. However, high concentration of HARCs (PE-0.1 % HARC) shows increase in crystallinity by ~ 3 %. One of the reasons for the latter case could be that randomly bent fibers and self entangled flocks of high aspect ratio CNTs cover the polymer. Further, their higher surface area to volume ratio (as compared to LARCs) increases the viscosity and nucleation sites for the crystallization in the composite and hence crystallinity improves. Further, HARCs are nicely embedded within matrix due to their morphological similarity with polymer chain and it is discussed in detail later in subsection 3.3. The higher specific heat of CNTs ($C_p$=0.75 $Jg^{-1}K^{-1}$ at 300 K [47]) can cause a heat pool in the surrounding polymer, which thermodynamically helps rearranging the polymer chains in a manner to increase the crystallinity. But 0.05 wt. % is not a sufficient filler fraction for giving such effect and hence PE-0.05HARC has shown almost no change in crystallinity. This observation is in line with the findings of other researchers [43, 48]. Therefore, aspect ratio and volume fraction of the CNTs influences the crystallinity behavior of the composite, which indirectly affects the mechanical properties.

Effect of CNT aspect ratio on thermal degradation behavior of the composites can be explained from the TGA plots presented in Fig.5. It is observed that high aspect ratio carbon nanotubes reduce the amount of oxidation or degradation of the composite at higher temperature. At 400° C, the mass remaining for PE, PE-0.05HARC and PE-0.1HARC is 81.13 %, 90.91 % and 92.45 % respectively. This is due to high aspect ratio and surface area of CNTs which are sufficient to cover the polymer particles and to give protection from degradation. Further, more carbon – carbon bonds in HARCs and their self entanglement do not allow the polymer to melt. In addition, HARC reinforcement also induces more crystallinity to the polymer matrix, as discussed before, which makes the structure more stable at higher temperature. As a result, HARC reinforced composite shows more stability at higher temperature as compared to pure polymer. In case of LARCs, the mass loss is much higher in composite than pure PE, with a 79 % remaining for PE-0.1LARC and 71 % for PE-0.05LARC at 400° C. It is due to the fact that uniformly dispersed stiff fibers expose their surface freely resulting in easy oxidation of amorphous carbon present in the CNTs and other carbonaceous impurity. Since the LARCs have much mismatch of size with that of polymer chains, there is a high chance their surfaces will be exposed and oxidized more than the HARCs, as the latter would be wrapped more by polymers. Further, LARC reinforcement decreases the crystallinity by disturbing the polymer chain arrangement, which leads to increase in amorphous region. Polymer chains in amorphous regions easily degrade when it is exposed to higher temperature. Hence, more mass loss is found in composite with less concentration of LARCs as compared to pure PE. But, increasing the concentration of LARCs reduces the mass loss to some extent, because more CNTs will stabilize the composite by covering polymer and also with increase in CNT content, the population of C-C bonds increases and as these bonds does not degrade easily at the operating temperature. These two becomes competing phenomenon, which causes some decrease in degradation rate at PE-0.1LARC as compared to PE-0.05LARC.

As a whole, it can be conceived that even low concentration of HARCs are also capable of significantly improving the thermal stability of polyethylene matrix.

### 3.3 Effect of CNT Aspect Ratio on Hardness and Elastic Modulus

The mechanical properties of the composites are evaluated through instrumented indentation technique. Fig.6 shows the representative load-displacement plots obtained from nano indentation test of different compositions with a depth controlled indentation mode. It can be observed that the load required to attain 150 nm depth of indent is higher for all the CNT reinforced PE composite. Presence of CNTs in polymer matrix helps in sharing load and thus the composites show higher hardness and elastic modulus. Fig.7 shows the average hardness and elastic modulus values and their standard deviations for at least 25 indents on each composition. Improvement in hardness and modulus is observed in the range of 3-45% and 8-42%, respectively, with increasing the concentration of different